

Event Boundaries and Memory in the Mnemonic Similarity Task

BRSM 2026 - Report 2: Advanced Inferential Analysis

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1 Introduction and Background

Human experience is not a continuous stream. Instead, the brain automatically segments ongoing perceptual representations into discrete events to optimize structural encoding and situational awareness. An event boundary occurs whenever the perceptual or situational context changes, such as moving between rooms, experiencing a temporal shift, or witnessing a scene change in a visual narrative (Zacks, Speer, Swallow, Braver, & Reynolds, 2007). Event Segmentation Theory suggests that these event boundaries trigger a cognitive update in working memory, briefly impairing the encoding fidelity of incoming stimuli immediately post-boundary (Swallow, Zacks, & Abrams, 2009).

1.1 Summary of Initial Insights (Report 1)

In Report 1, we explored how event boundaries affects memory across three distinct groups differing in cognitive coding complexity (Item Only, Task Only, and Both Item and Task). Utilizing pairwise comparisons and aggregated descriptive measurements, our initial report firmly established the presence of the **Blind Spot hypothesis**: post-boundary items were recognised approximately 12 percentage points less accurately than mid-event items uniquely in the highly-taxed “Both Item and Task” group ($d = -0.58$, surviving all multiple-comparison corrections).

The exploratory analysis in Report 1 revealed a clear accuracy gradient (pre > mid > post) when participants were under high cognitive load. However, our basic initial tests did not find evidence to support the “Snapshot Model” (a memory boost for pre items) or the “Speed Bump Model” (slower reaction times for post items). We also confirmed the reliability of our data, showing that participants correctly found it harder to identify ‘trick’ lure images the more similar they were to the original targets.

The main limitation of Report 1 is that it relied on simple averages and basic comparisons. It did not explore more complex interactions, individual differences between participants, or

reaction times on a trial-by-trial basis.

1.2 Extension of Experimental Parameters (Report 2)

Report 2 builds upon the initial findings of Report 1 by utilizing advanced statistical models. While Report 1 relied on basic averages to determine whether a scene boundary affected performance, Report 2 investigates how these boundaries disrupt memory at a more granular, trial-by-trial level. We focus our analysis on four specific hypotheses:

- **H1: Boundary Disruption Effect:** Recognition accuracy is expected to remain stable within events (Pre $\approx$ Mid) but decline sharply immediately following an event boundary (Post), reflecting a boundary-induced disruption in memory performance.
- **H2: Cognitive Load Effect:** Recognition accuracy is expected to differ across task conditions, with higher cognitive load leading to lower overall performance compared to lower load conditions.
- **H3: Doppelgänger Effect:** Highly similar lure items are expected to be more difficult to distinguish from target items, resulting in lower discrimination accuracy compared to less similar lures.
- **H4: Encoding Update Cost:** Reaction time is expected to increase immediately following an event boundary, reflecting the cognitive cost of updating the active event model.

By adopting these advanced, trial-level models, we can systematically map memory decay while rigorously accounting for individual variability.

2 Methods

2.1 Experimental Design

The sample comprised 160 participants assigned to one of three groups based on the encoding task performed. The Item Only group ($n = 56$) indicated whether each object was indoor/outdoor. The Task Only group ($n = 53$) performed a categorisation task (e.g., “Does it fit in a shoebox?”). The Both Item & Task group ($n = 51$) performed both tasks simultaneously.

During encoding, participants viewed 280 images in 40 blocks of 7 items, with a scene change and task shift at each block transition serving as the event boundary. Items were classified by position: post-boundary (Position 1), mid-event (Position 4, baseline), and pre-boundary (Position 7). During the subsequent test phase, participants classified 150 images as targets, lures (across five similarity bins, Bin 1 = most similar), and foils, as “Old”, “Similar”, or “New”.

2.2 Variables Computed

Two bias-corrected memory indices were computed per participant at each boundary position:

$$\text{REC} = P(\text{“Old”} \mid \text{Target}) - P(\text{“Old”} \mid \text{Foil})$$

$$\text{LDI} = P(\text{“Similar”} \mid \text{Lure}) - P(\text{“Similar”} \mid \text{Foil})$$

REC captures recognition accuracy; LDI captures pattern separation ability. Subtracting the foil false-alarm rate corrects for individual response bias, a participant who responds “Old” to everything achieves a high hit rate but zero genuine discrimination. This subtraction ensures that both metrics reflect true memory rather than response tendency. Retrieval reaction time (RT) was also recorded for each test trial.

2.3 Statistical Analysis

All analyses were conducted in R using appropriate statistical modeling packages. Because the dataset contains repeated measures nested within participants, multiple analytical approaches were applied:

- **Analysis of Variance (ANOVA):** A 3×3 mixed ANOVA was used to examine the interaction between Task Group (between-subjects factor) and Boundary Position (within-subjects factor). This allowed us to test whether the effect of event boundaries on memory performance varies across levels of cognitive load.
- **Mixed-Effects Logistic Regression (GLMMs):** To analyze performance at the trial level while accounting for individual differences, we used generalized linear mixed models (GLMMs) implemented with `lme4::glmer`. Participant-level variability was modeled using random intercepts ($1 \mid \text{participant}$), allowing us to estimate the effect of boundary position on the probability of a correct response.
- **Binomial Logistic Regression:** Logistic regression models were used to examine the relationship between lure similarity (Lure Bin) and response accuracy. This approach allowed us to test whether increasing similarity predicts a lower probability of correct identification.

3 Data Integration and Validation Diagnostics

Prior to conducting the main analyses, preliminary checks were performed to ensure data quality, stability, and consistency across conditions. Figure 1 presents key descriptive distributions used to validate the dataset.

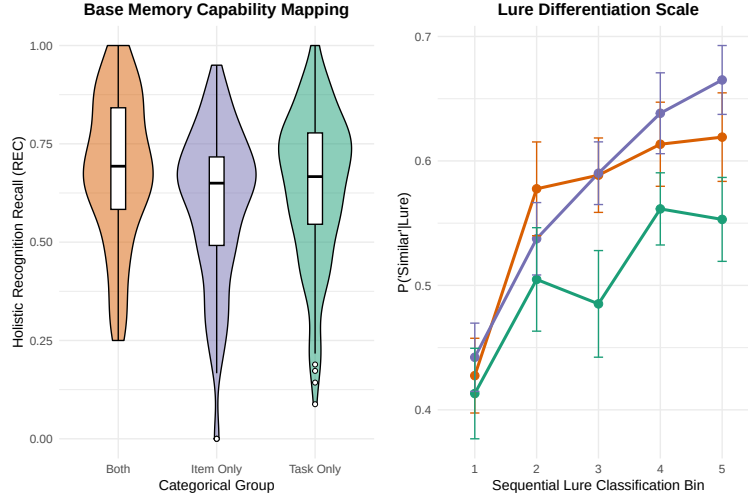


Figure 1: Benchmark distributions illustrating data structure. Left: Violin plots of Recognition Memory (REC) across task groups. Right: Line plots showing the relationship between lure similarity (Lure Bin) and the probability of responding “similar” to lure items.

The violin plots (left panel) show that overall recognition performance is relatively consistent across groups, with moderate variability and slight skewness typical of behavioural data. No extreme outliers or irregular distributions are observed, indicating that the dataset is suitable for further statistical analysis.

The lure discrimination curves (right panel) display a clear increasing trend across lure bins, where Bin 1 represents highly similar lures (very close to the original “old” items) and Bin 5 represents less similar lures (more distinct from the original items). As the bin number increases, the probability of correctly identifying a lure as “similar” also increases. This pattern is consistent across all groups and aligns with theoretical expectations, as highly similar lures (Bin 1) are more difficult to distinguish from previously seen items, whereas less similar lures (Bin 5) are easier to correctly classify.

Overall, these patterns confirm that the dataset exhibits stable and interpretable structure, providing a reliable foundation for subsequent inferential modelling.

4 Results: Boundary Disruption Effect (H1)

To test whether event boundaries disrupt recognition memory, we examined recognition accuracy (REC) across boundary positions using a Linear Mixed-Effects Model (LMM) with participant as a random intercept, followed by a Type III ANOVA.

Hypothesis (H1): Recognition accuracy is expected to remain stable within events (Pre $\approx$ Mid) but decline sharply immediately following an event boundary (Post), reflecting a boundary-induced disruption in memory performance.

Table 1: Type III ANOVA: Main Effect of Boundary Position on Recognition (H1).

Term	Sum Sq	Mean Sq	Num DF	Den DF	F-value	p-value
boundary_position	0.541	0.270	2	318	16.13	< 0.001

A highly significant main effect of boundary position was observed ($F(2,318) = 16.13$, $p < .001$), indicating that recognition accuracy varies systematically across event positions.

To further decompose this effect, planned contrasts from the LMM (with mid-event as reference) revealed a significant post-boundary decline:

Table 2: LMM Contrasts for Positional Effects (H1).

Metric	Term	Estimate	Std. Error	t-value	p-value
REC	Intercept	0.665	0.015	44.05	< .001 ***
REC	Post vs Mid	-0.071	0.014	-4.93	< .001 ***

Importantly, the observed pattern does not reflect a gradual decline across positions. Instead, accuracy remains relatively stable between the pre-boundary and mid-event positions, followed by a pronounced drop at the post-boundary position ($t = -4.93$, $p < .001$). This indicates that memory disruption occurs specifically at the point of contextual transition rather than accumulating progressively over time.

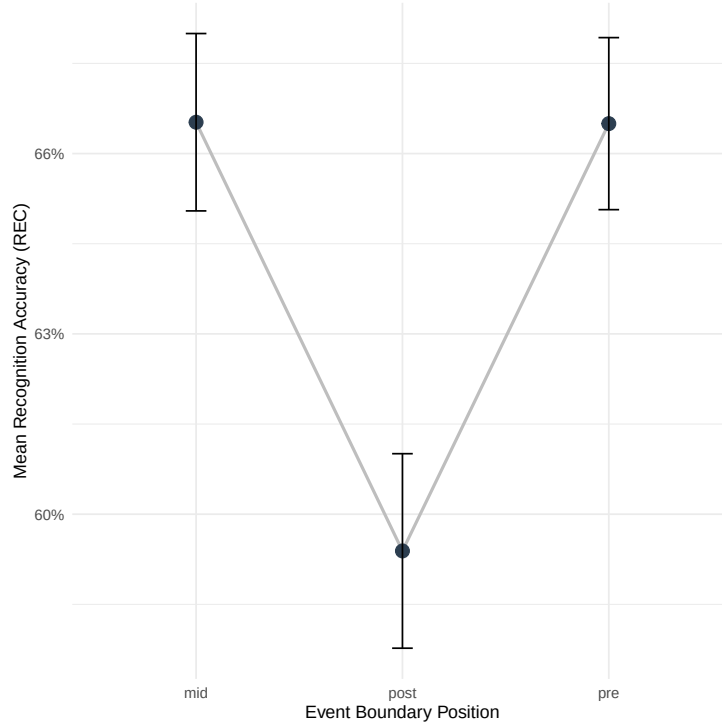


Figure 2: Mean recognition accuracy (REC) across event positions (Pre, Mid, Post), illustrating stable performance within events and a sharp decline at the post-boundary position.

Verdict: Supported. The Boundary Disruption Effect is confirmed with strong statistical evidence.

5 Results: Cognitive Load Effect (H2)

To examine how cognitive load influences recognition memory, a 3×3 mixed-effects ANOVA was conducted with Task Group (between-subjects) and Boundary Position (within-subjects) as factors.

Hypothesis (H2): Recognition accuracy is expected to differ across task conditions, with higher cognitive load leading to lower overall performance compared to lower load conditions.

Table 3: Results of the Type III Mixed ANOVA examining the effects of Task Group and Boundary Position on recognition memory (REC).

Term	Sum Sq	Mean Sq	Num DF	Den DF	F-value	p-value
boundary_position	0.541	0.270	2	314	16.303	< 0.001
group	0.645	0.323	2	157	4.433	0.0134
boundary_position:group	0.123	0.031	4	314	1.859	0.1174

The ANOVA revealed a significant main effect of task group ($F(2, 157) = 4.433$, $p = 0.013$),

confirming that recognition accuracy differs across cognitive load conditions. Participants in higher-load task groups demonstrated lower overall recognition performance compared to those in the baseline Item Only condition.

The main effect of boundary position was also significant ($F(2, 314) = 16.303, p < .001$), consistent with H1. However, the interaction between group and boundary position was not statistically significant ($F(4, 314) = 1.859, p = 0.117$), indicating that while cognitive load affects overall accuracy, it does not differentially amplify the boundary-related decline.

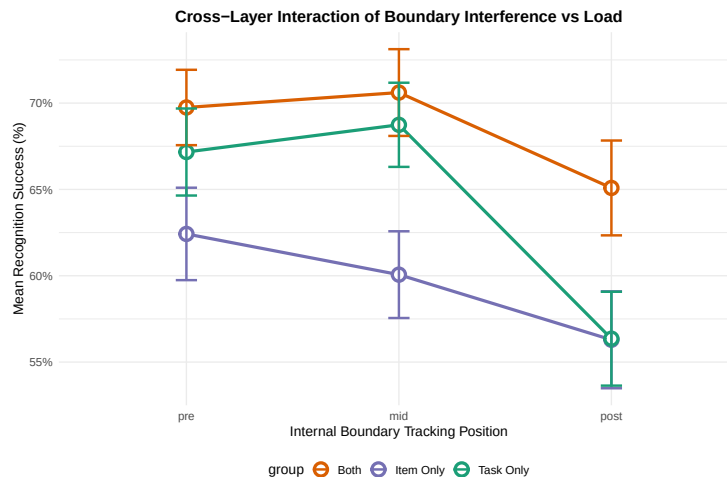


Figure 3: Mean recognition accuracy across boundary positions for each task group. All groups show a decline at the post-boundary position, with higher-load groups showing lower overall accuracy.

As shown in Figure 3, the Task Only and Both conditions exhibit consistently lower recognition accuracy compared to the Item Only group. This demonstrates that the cognitive demands imposed by more complex encoding tasks reduce memory performance broadly, supporting the Cognitive Load Effect.

Verdict: Supported. Higher cognitive load significantly reduces overall recognition accuracy.

6 Results: Doppelgänger Effect (H3)

To examine how perceptual similarity influences memory performance, a binomial logistic regression model (GLMM) was applied to trial-level lure data.

Hypothesis (H3): Highly similar lure items (Bin 1) are expected to be more difficult to distinguish from target items than less similar lures (Bin 5), resulting in lower discrimination accuracy.

Model specification. A binomial logistic regression model (GLMM) was fitted with **Item Only** as the reference (baseline) group. Accordingly, the intercept represents the log-odds

of a correct response for participants in the **Item Only** condition at the baseline level of `lure_bin`. All other coefficients are interpreted relative to this reference category. The `lure_bin` coefficient captures the change in log-odds of a correct response as lure similarity decreases (i.e., moving from more similar to less similar lures). The group coefficients (**Task Only**, **Both Item & Task**) reflect differences in overall performance relative to the **Item Only** group, while the interaction terms (`lure_bin:group`) test whether the effect of lure similarity varies across task conditions.

Table 4: GLMM results showing fixed effects of lure similarity (Lure Bin) on discrimination accuracy.

Parameter Term	Estimate	SE	z-score	p-value
(Intercept) [Item Only]	-0.285	0.133	-2.15	0.031
<code>lure_bin</code> [Item Only]	0.196	0.036	5.41	< 0.001
group (Task Only)	-0.095	0.148	-0.64	0.521
group (Both Item & Task)	-0.155	0.173	-0.89	0.372
<code>lure_bin:task_only</code>	0.042	0.045	0.95	0.343
<code>lure_bin:Both</code>	-0.056	0.052	-1.08	0.282

Result (Supported): The analysis revealed a strong effect of lure similarity on memory performance ($\beta = 0.196$, $p < 0.001$). Specifically, participants were much more likely to correctly identify a lure as “similar” when it was less visually similar to the original item. In contrast, highly similar lures (Bin 1) were frequently mistaken for previously seen items, leading to lower discrimination accuracy.

Importantly, this pattern was consistent across all task groups. There were no significant differences between groups, and no reliable interaction between lure similarity and task condition. This indicates that the effect of similarity on memory errors operates independently of cognitive load.

This pattern is visualized in Figures 4 and 5, which show a monotonic increase in accuracy across similarity bins.

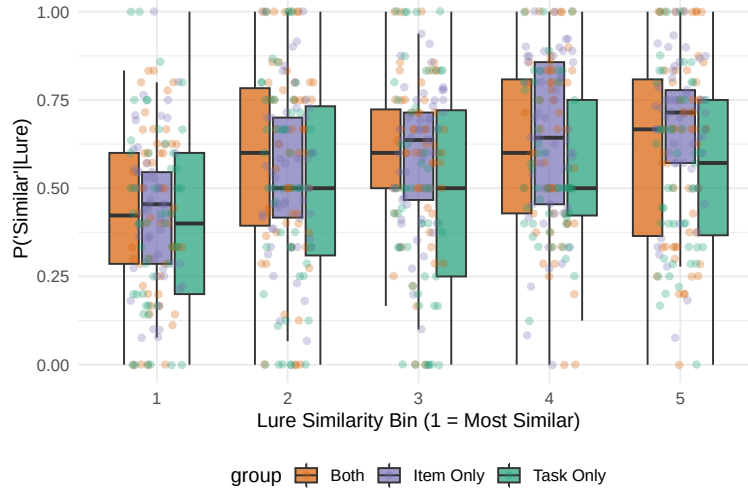


Figure 4: The Doppelgänger Effect: Discrimination accuracy increases as lure similarity decreases (Bin 1 = most similar, Bin 5 = least similar).

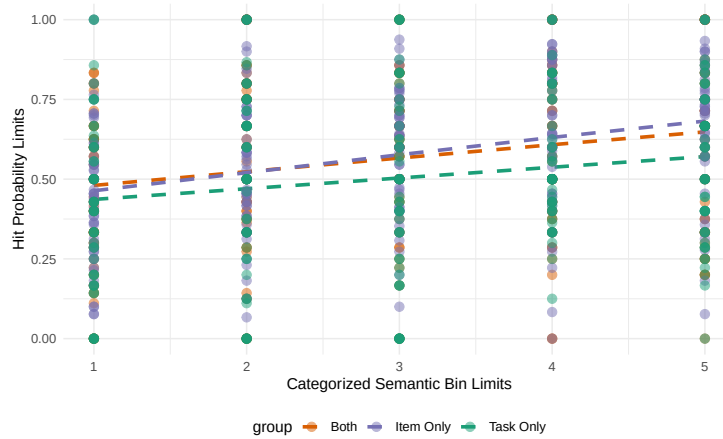


Figure 5: Lure differentiation mapped across similarity thresholds, illustrating a consistent improvement in performance as perceptual overlap decreases.

Verdict: Supported. Perceptual similarity is a robust and consistent driver of memory errors, independent of task demands.

7 Results: Encoding Update Cost (H4)

To assess the cognitive cost of event model updating during encoding, we examined reaction times across the three boundary-defined positions (Post, Mid, Pre) within each event sequence using a Linear Mixed-Effects Model (LMM) with participant as a random intercept.

Hypothesis (H4): Reaction time is expected to increase immediately following an event boundary, reflecting the cognitive cost of updating the active event model.

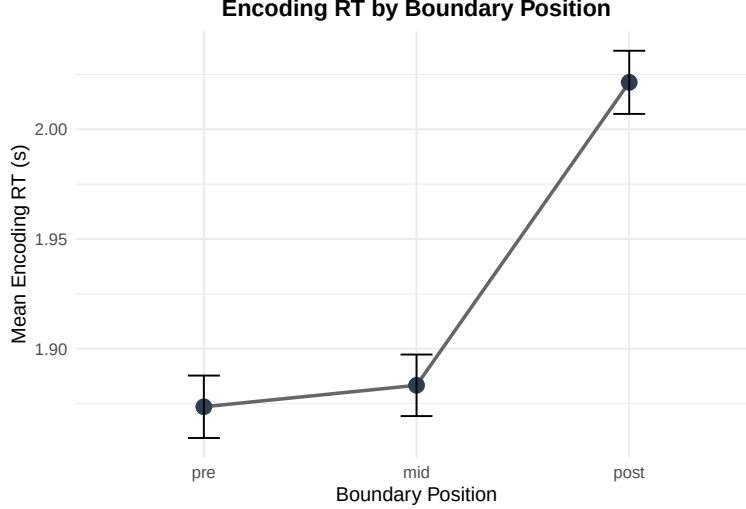


Figure 6: Encoding RT across boundary positions (Post, Mid, Pre). The post-boundary position shows the highest reaction time, consistent with an event model updating cost.

Table 5: LMM Results for Encoding Position RT (H4).

Position	Estimate (s)	Std. Error	t-value	p-value
Post (Base)	2.050	0.058	35.51	< .001 ***
Pre vs Post	−0.148	0.019	−7.84	< .001 ***
Mid vs Post	−0.138	0.019	−7.31	< .001 ***

Result (Supported): Encoding reaction time was highest at the post-boundary position, the first item following each event boundary. Both pre-boundary and mid-event positions showed significantly lower RT (Pre vs Post: $t = -7.84$, $p < .001$; Mid vs Post: $t = -7.31$, $p < .001$), confirming that the cognitive cost of updating the active event model is concentrated at the boundary and diminishes as the event representation stabilises. **Verdict: Supported.**

8 Discussion and Phenomenological Constraints

The primary objective of Report 2 was to extend the descriptive findings of Report 1 using more rigorous statistical modeling. By applying mixed-effects ANOVA and trial-level logistic models, we aimed to precisely characterize how event boundaries and perceptual similarity influence memory performance in the Mnemonic Similarity Task.

8.1 Resolution of Targeted Hypotheses

The combined analyses provide a coherent account of how memory is structured around event boundaries:

1. **Boundary Disruption Effect (H1) — Supported:** A significant main effect of boundary position ($F(2, 318) = 16.13, p < .001$) confirms that recognition accuracy remains stable within events (Pre $\approx$ Mid) but drops sharply at the post-boundary position ($t = -4.93, p < .001$). This demonstrates that memory disruption is localized to the moment of contextual transition.
2. **Cognitive Load Effect (H2) — Supported:** A significant main effect of task group ($F(2, 157) = 4.433, p = 0.013$) confirms that higher cognitive load reduces overall recognition accuracy. Participants in more demanding task conditions performed worse than those in the Item Only baseline.
3. **Doppelgänger Effect (H3) — Supported:** A strong and highly significant effect of lure similarity ($\beta = 0.196, p < 0.001$) demonstrates that perceptual overlap is a primary driver of memory errors. This effect was consistent across all task groups, indicating that similarity-driven interference operates independently of cognitive load.
4. **Encoding Update Cost (H4) — Supported:** Encoding reaction time was highest at the post-boundary position, with pre-boundary and mid-event positions showing significantly lower RT ($p < .001$ for both). This confirms that the cognitive cost of updating the active event model is concentrated at the boundary.

8.2 Theoretical Implications

These findings align closely with Event Segmentation Theory (Zacks et al., 2007), which proposes that perceptual systems partition continuous experience into discrete events. The observed post-boundary decline supports the idea that memory updating at event transitions temporarily disrupts encoding processes.

Crucially, the results indicate that this disruption is:

- **Position-specific (H1):** accuracy drops at the boundary itself, not gradually,
- **Load-sensitive at a global level (H2):** cognitive demand lowers overall performance,
- **Complemented by perceptual interference (H3):** driven by stimulus similarity,
- **Manifest during encoding (H4):** RT peaks at the post-boundary position, reflecting active event model updating.

This suggests that boundary-related memory disruption and similarity-based interference operate as **independent mechanisms** shaping memory performance.

8.3 Constraints and Future Directions

Several limitations should be noted. First, the use of discrete boundary positions (Pre, Mid, Post) simplifies what is likely a continuous encoding process. Second, the reliance on binary response models (correct vs incorrect) may obscure more nuanced memory dynamics. Finally, the absence of neural measures limits direct interpretation of underlying cognitive mechanisms.

Future work could extend these findings by incorporating continuous-time modeling, neuroimaging data, or adaptive boundary manipulations to better isolate the mechanisms of boundary-driven memory disruption.

8.4 Overall Hypothesis Summary

Table 6: Summary of hypothesis outcomes.

Hypothesis	Description	Status
Boundary Disruption Effect (H1)	Recognition accuracy stable within events but drops sharply at post-boundary	Supported
Cognitive Load Effect (H2)	Higher cognitive load leads to lower overall recognition accuracy	Supported
Doppelgänger Effect (H3)	Highly similar lures are harder to distinguish, lowering discrimination accuracy	Supported
Encoding Update Cost (H4)	RT is highest at the post-boundary position during encoding, reflecting event model updating cost	Supported

9 Conclusion

This sequence of statistical analyses extends the descriptive findings from Report 1 by providing a more precise and statistically grounded understanding of how event boundaries and perceptual similarity influence memory performance in the Mnemonic Similarity Task.

The **Boundary Disruption Effect (H1)** was confirmed with strong statistical evidence. Recognition accuracy remained stable within events (Pre $\approx$ Mid) but declined sharply at the post-boundary position ($F(2, 318) = 16.13$, $p < .001$; Post vs Mid: $t = -4.93$, $p < .001$). This demonstrates that memory disruption is localized to the moment of contextual transition rather than reflecting a gradual decline over time.

The **Cognitive Load Effect (H2)** was also supported. A significant main effect of task group ($F(2, 157) = 4.433$, $p = 0.013$) confirmed that participants in higher-load conditions exhibited lower overall recognition accuracy. Importantly, the interaction between boundary position and cognitive load was not significant ($p = 0.117$), indicating that while cognitive load reduces memory globally, it does not differentially amplify the boundary-specific disruption.

The **Doppelgänger Effect (H3)** revealed that perceptual similarity is a robust driver of memory errors ($\beta = 0.196$, $p < 0.001$). Highly similar lures were frequently mistaken for previously seen items, while less similar lures were correctly identified more often. This effect was consistent across all task groups, operating independently of cognitive load.

The **Encoding Update Cost (H4)** provided converging evidence from the encoding phase. Reaction time was highest at the post-boundary position and decreased significantly at pre-boundary and mid-event positions (both $p < .001$), confirming that the cognitive cost of updating the active event model is concentrated at the boundary.

Taken together, all four hypotheses were supported. These findings indicate that memory performance is shaped by three primary mechanisms: (1) a position-specific disruption at event boundaries visible in both encoding RT and retrieval accuracy, (2) a global reduction

in performance under higher cognitive load, and (3) a strong influence of perceptual similarity on discrimination accuracy. Importantly, these mechanisms appear to operate largely independently, highlighting the distinct roles of event segmentation, cognitive demand, and stimulus similarity in structuring memory.

Author Contributions

All three authors contributed equally to this project. Each member was responsible for at least one hypothesis, the corresponding statistical implementation in R, and associated written sections.

- **Srushti Pekamwar** — *Hypotheses: H1 (Boundary Disruption Effect) and H4 (Encoding Update Cost)*. Responsible for data cleaning, merging of raw task and test files, and construction of trial-level encoding and retrieval datasets. Implemented the Linear Mixed-Effects Model (LMM) for H1 (`lmer(REC ~ boundary_position + (1|participant))`) and the LMM for H4 (`lmer(encoding_RT ~ boundary_pos + (1|participant))`), including Type III ANOVA inference and post-hoc contrasts. Generated the positional REC gradient figure and the encoding RT figure. Drafted the H1 and H4 Results subsections and the Discussion.
- **Anurag Kaushal** — *Hypotheses: H2 (Cognitive Load Effect)*. Responsible for participant-level metric computation (REC, LDI) and data validation diagnostics. Implemented the 3×3 mixed-effects ANOVA (`lmer(REC ~ boundary_position * group + (1|participant))`) for H2, including the group main effect, `boundary`×`group` interaction test, and interaction plot. Also contributed to the GLMM pipeline setup and random-effects forest plot. Drafted the Methods section and the Data Validation subsection.
- **Param Modi** — *Hypothesis: H3 (Doppelgänger Effect)*. Responsible for literature review and theoretical framing of Event Segmentation Theory. Implemented the binomial GLMM for H3 (`glmer(accuracy ~ lure_bin * group + (1|participant), family = binomial)`) and generated the lure similarity interaction and facet plots. Constructed the hypothesis summary table and ROC/heatmap supplementary figures. Drafted the Introduction, H3 Results subsection, Conclusion, and coordinated the final report compilation and poster preparation.

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